$$X(2,6) = X(3,5) = X(4,4) = X(5,3) = X(6,2) = 8$$

$$X(3,6) = X(4,5) = X(5,4) = X(6,3) = 9$$

$$X(4,6) = X(5,5) = X(6,4) = 10$$

$$X(5,6) = (6,5) = 11$$

$$X(6,6) = 12.$$

(ii) Then the random variable $X$ takes on the values 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

(iii) The inverse images of 10 is $\{(4, 6), (5, 5), (6, 4)\}$.

(iv) The number of inverse images are given below

| Values of the random variable | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of elements in inverse image | 1 | 2 | 3 | 4 | 5 | 6 | 5 | 4 | 3 | 2 | 1 | 36 |

**Example 11.3**

An urn contains 2 white balls and 3 red balls. A sample of 3 balls are chosen at random from the urn. If $X$ denotes the number of red balls chosen, find the values taken by the random variable $X$ and its number of inverse images.

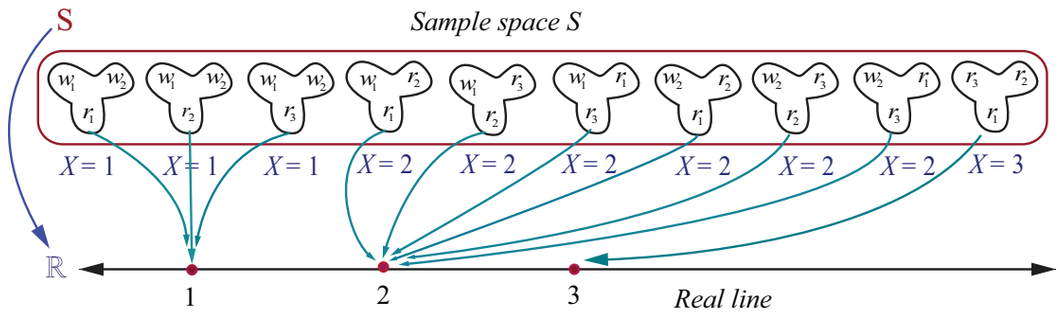
**Solution**

Let us denote white and red balls as $w_1, w_2, r_1, r_2,$ and $r_3$.

The sample space consists of $5c_3 = 10$ different samples of size 3.

That is $S = \{w_1w_2r_1,\ w_1w_2r_2,\ w_1w_2r_3,\ w_1r_1r_2,\ w_1r_2r_3,\ w_1r_1r_3,\ w_2r_1r_2,\ w_2r_2r_3,\ w_2r_1r_3,\ r_1r_2r_3\}$.

The random variable $X$ takes on the values 1, 2, and 3.




**A mapping $X$ (.) from $S$ to real numbers**

Fig. 11.3

| Values of the Random Variable $X$ | 1 | 2 | 3 | Total |
|---|---|---|---|---|
| Number of elements in inverse images | 3 | 6 | 1 | 10 |

**Remark**

If $X$ denotes the number of white balls, then $X$ takes on the values 0,1, and 2 and the elements in inverse images are



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